João André **Roque Costa**

MSc Graduate, Computer Science and Engineering

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Computer Science professional with a Master's degree in Computer Science (AI and Data Science) from **Instituto Superior Técnico**. Experienced in **Machine Learning**, **Deep Learning**, and **Data Science**, with a thesis focused on scalable nonlinear dimensionality reduction. Proficient in **Python**, **SQL**, and applied analytics, with hands-on work across data analysis, automation, and stakeholder-facing AI projects.

Professional Experience

Deloitte — *Coimbra, Portugal*

Tech Consultant — *Apr. 2026 - Oct. 2026*

- **Planned** and **developed** a multi-agent application for data migration and synthetic data generation (Claude SDK, Azure Blob Storage, MongoDB).
- **Ported** and updated an Azure resource group between two subscriptions, coordinating Azure Managed Instances, Azure Data Factory, and Databricks.
- **Automated** an HR report generation workflow for recurring stakeholder reporting using Python and Playwright.
- **Drafted** a client proposal to automate documentation processes with RAG and OCR.

Instituto Superior Técnico — *Lisbon, Portugal*

Invited Teaching Assistant — *Oct. 2024 - Jul. 2025*

- **Lectured** and **evaluated** practical classes on Databases and Data Science.
- **Supported** students with hands-on projects in data treatment, visualization, and analysis using Python (pandas, matplotlib), reinforcing structured communication of technical results.

Education

Instituto Superior Técnico — *Lisbon, Portugal*

M.Sc. Computer Science and Engineering — *Sep 2023 - Nov 2025*

- Specialized in **Artificial Intelligence** with emphasis in **Engineering and Data Science**, combining theory with practical applications in modeling and data-driven problem solving.
- **Relevant Coursework**: Optimization Algorithms, Deep Learning, Cloud Computing and Virtualization, Machine Learning.
- Thesis: **Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)**.
  - Developed a novel extension to the MVU algorithm that enables **nonlinear dimensionality reduction** on disconnected datasets.
  - Engineered a decomposition-based approach that **allows for parallel processing**, improving scalability for large-scale datasets that were previously computationally infeasible.
  - Preserved local geometric structure while achieving embeddings comparable to the standard method.

  Engenharia e Ciência de Dados / Data Science

Instituto Superior Técnico — *Lisbon, Portugal*

B.Sc. Computer Science and Engineering — *Sep 2020 - Jul 2023*

- **Built** a strong foundation in Analysis and Synthesis of Algorithms, Distributed Systems, Operating Systems, and Database Theory.
- **Worked** on full-stack and systems-level projects with emphasis on **memory management** in **C/C++** and relational schema design in **SQL**.

Extracurricular

Personal Software Projects (Self-Hosted)

- Engineered an automated betting bot for parimutuel systems using a modified Kelly Criterion for **risk management** and **optimal stake sizing**.
- Developed a custom Discord-based ELO ranking and **automatic matchmaking system**.
- Architected a multi-platform **arbitrage engine** and inventory manager. With automated price monitoring and trade execution across multiple marketplaces.
- Built a full-stack system for **managing multiple modular projects** in a linux nuc (per machine resource management, product price tracking).

Skills & Competencies

Data Science	**Machine Learning**, **Deep Learning**, **LLMs**, statistical analysis, predictive modeling, data visualization
Programming	**Python** (Claude-Agent-SDK, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Playwright)
Data & Databases	**SQL**, PostgreSQL, Microsoft SQL Server (SSMS), MongoDB, MySQL, pgAdmin
Data Engineering	Databricks, **Pentaho PDI**, SQL Profiler, Tuning Advisor (DTA), DataCleaner tool
Cloud & Architecture	**Azure**, **AWS**, ArchiMate
Developer Tools	Git, Linux, Bash Scripting, RESTful APIs, LaTeX
Languages	Portuguese (Native), English (Fluent)